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DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

FINAL YEAR PROJECT

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PROJECT TITLE : BLACK SPOT DETECTION USING MACHINE LEARNING(GPS)
ROLL NUMBER : 24NU1F0048
DEPARTMENT : MCA
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ACADEMIC YEAR : 2024 - 2026

OUTLINE

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INTRODUCTION

The **Black Spot Detection Using Machine Learning (GPS)** system uses historical accident data, GPS coordinates, and machine learning algorithms to identify and predict accident-prone locations. The system analyzes accident patterns, classifies high-risk areas, and displays them on digital maps for easy visualization. GPS technology enables accurate location tracking, while machine learning improves prediction accuracy based on past data.

PROBLEM STATEMENT

Road accidents are increasing rapidly, and many of them repeatedly occur at specific locations called black spots. Existing accident analysis systems mainly depend on manual surveys and historical reports, which are time-consuming and not suitable for real-time usage. Drivers are usually unaware of these dangerous zones until an accident happens. Current navigation systems do not provide proactive warnings about accident-prone areas. There is no intelligent mechanism to predict or classify black spots using data-driven techniques. Hence, an automated and intelligent system is required to identify, predict, and alert users about black spots using Machine Learning and GPS technology.

OBJECTIVES

1. To analyze historical accident data and identify black spots using machine learning algorithms.
2. To classify road locations based on accident frequency and severity.
3. To provide real-time alerts to users when they approach accident-prone zones.
4. To visualize black spots on an interactive map for better understanding.
5. To assist transport authorities in decision-making through analytical insights.

PROPOSED SYSTEM

Proposed System (Step-by-Step)

Step-1: Collect historical road accident dataset

Step-2: Preprocess data (cleaning, normalization)

Step-3: Train machine learning model

Step-4: Classify locations as black spot or non-black spot

Step-5: Integrate GPS for real-time location tracking

Step-6: Display black spots on interactive map

Step-7: Generate alerts when user approaches a black spot

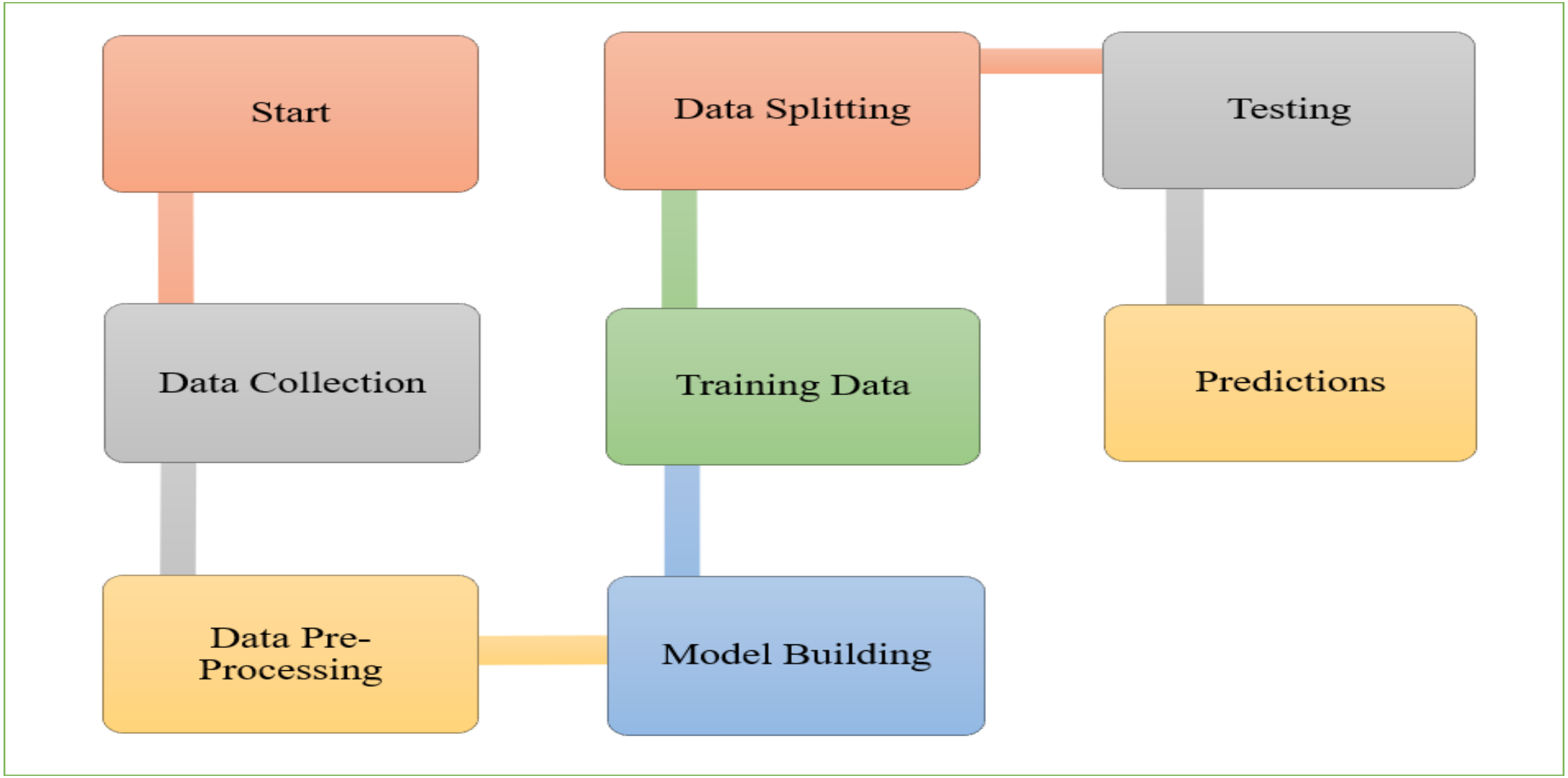
LITERATURE SURVEY

S.No	Author & Year	Title	Description	Links
1	Abdel-Aty et al., 2011	Accident analysis using ML	A machine learning method to study past road accident data.	https://doi.org/10.1016/j.aap.2011.01.001
2	Chang & Wang, 2006	Crash prediction models	Their method improved accident prediction accuracy compared to traditional techniques.	https://doi.org/10.1016/j.trc.2006.05.002
3	Xie et al., 2014	Road safety analysis	SVM (Support Vector Machine) technique to study road accident data.	https://doi.org/10.1016/j.ssci.2014.01.007
4	Kumar & Toshniwal, 2016	Accident severity prediction	The Random Forest technique to predict accident severity.	https://ieeexplore.ieee.org
5	Kwon et al., 2015	GIS-based accident analysis	Their approach helped in finding accident-prone areas by location (spatial analysis).	https://doi.org/10.1016/j.jtrangeo.2015.02.004
6	Chen et al., 2019	Traffic accident prediction	Deep Learning techniques to predict traffic accidents	https://doi.org/10.3390/s19092103
7	Pande & Abdel-Aty, 2009	Crash risk estimation	Their approach helped in real-time accident risk analysis and improved roads	https://doi.org/10.1016/j.aap.2009.01.006

TOOLS & TECHNOLOGIES

Layer	Tools & Technologies
Frontend	HTML, CSS, JavaScript, Google Maps API
Backend	Python, Flask, ML algorithms
Dataset	Historical accident data (CSV)
Database	MySQL / SQLite
Programming Language	Python
Machine Learning Algorithms	Random Forest, Supervised Learning Models
Libraries	NumPy, Pandas, Scikit-learn
Mapping & GPS	OpenStreetMap / Google Maps
Development Tools	Python IDE, Machine Learning Frameworks

SYSTEM ARCHITECTURE



METHODOLOGY

Data Collection: Accident data is collected from traffic records containing location, time, severity, and cause.

Data Preprocessing: Missing values are removed and features are normalized for better model performance.

Model Training: Machine learning algorithms such as Random Forest or SVM are trained using historical data.

Prediction: The trained model predicts whether a road segment is accident-prone.

GPS Integration: User's real-time location is tracked using GPS.

Alert System: When the user is near a black spot, voice and visual alerts are generated.

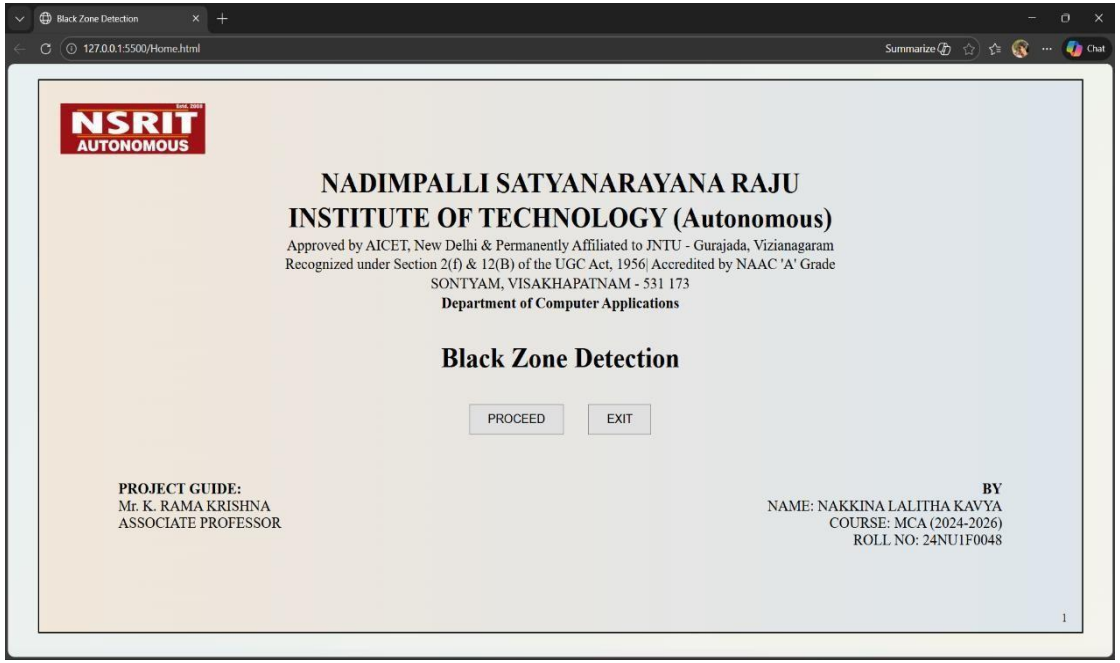
Visualization: Black spots are highlighted on maps for easy understanding.

RESULTS

MODULE-WISE IMPLEMENTATION

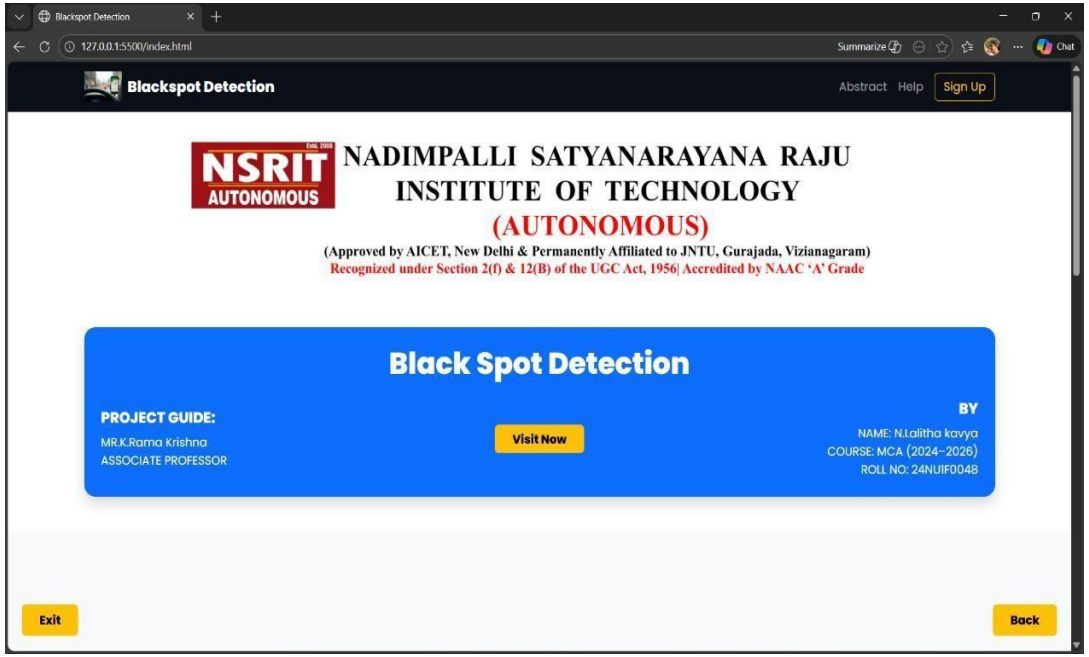
MODULE – 1 :

Module Status: Completed



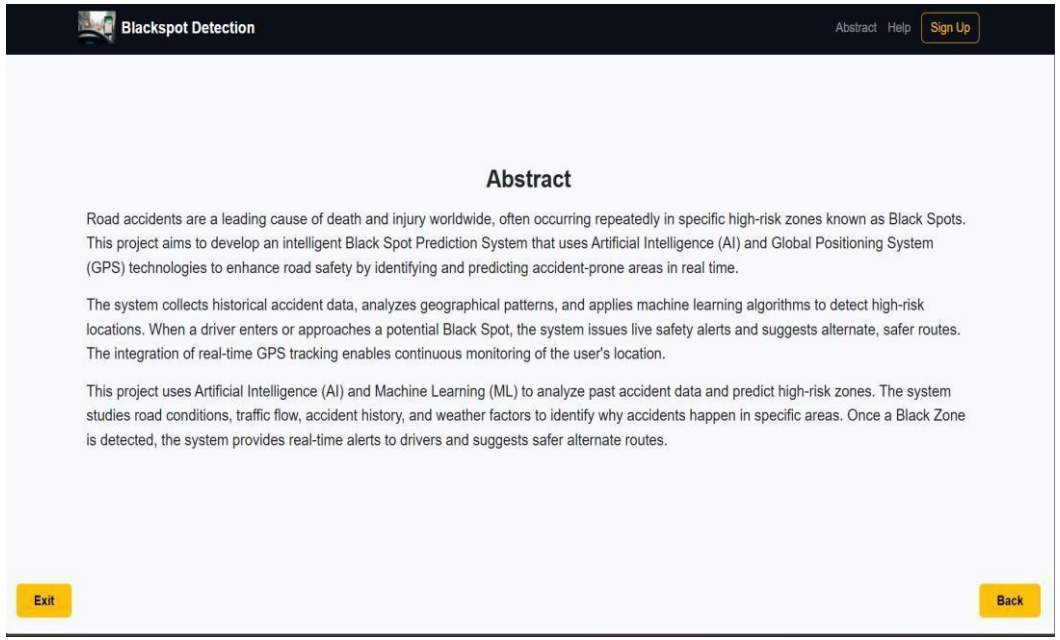
HOME PAGE

Project Execution starts from the welcome page having the title of the project. If the user click on "Proceed" button then it will take user into the menu page.



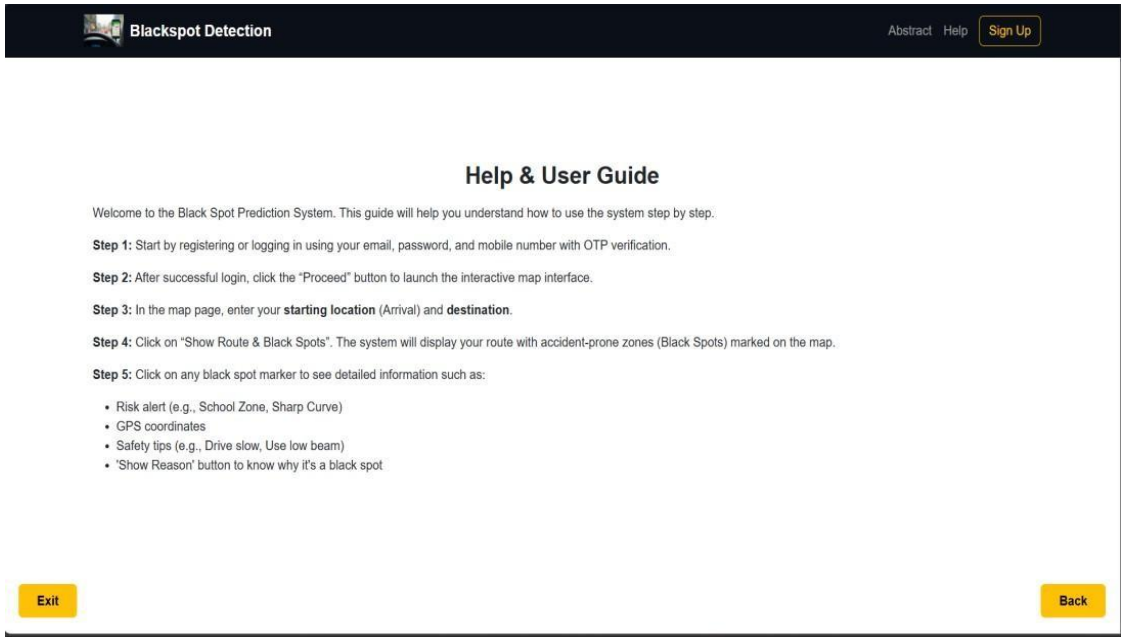
MENU PAGE

If the user clicks on "Visit" button then it will display the login page of the application.



ABSTRACT PAGE

If the user clicks on “Abstract” button then it will Scroll downs the Abstract page

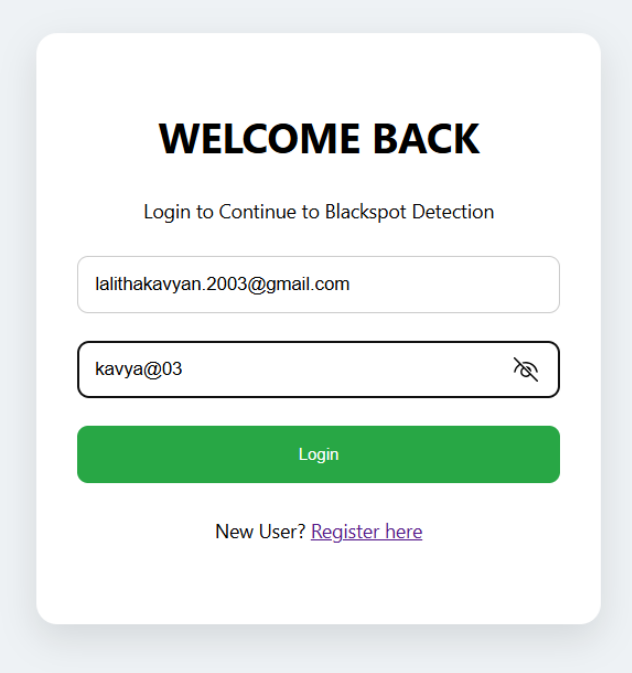


HELP& USER GUIDE PAGE

If the user clicks on “Help” button then it will Scroll downs the Help page.

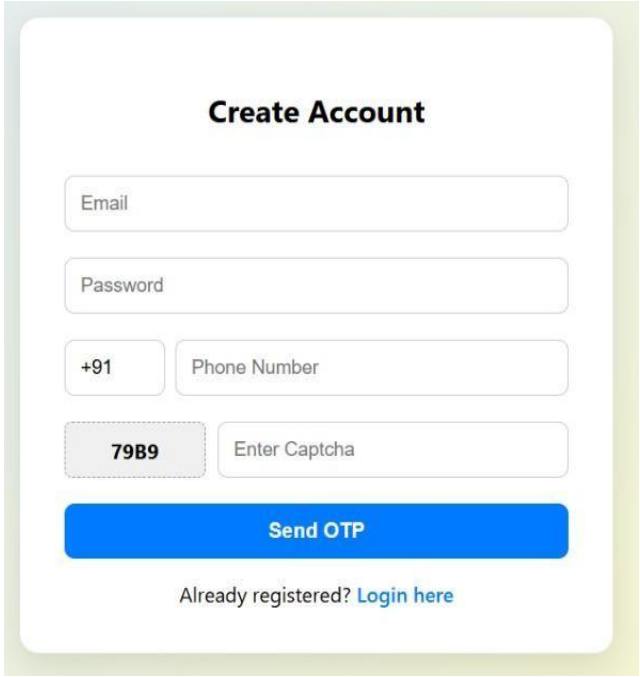
MODULE – 2 :

Module Status: Completed



LOGIN PAGE

On Clicking “Login Button” then it redirects to the Login page and enter them personal details already given while Registered.

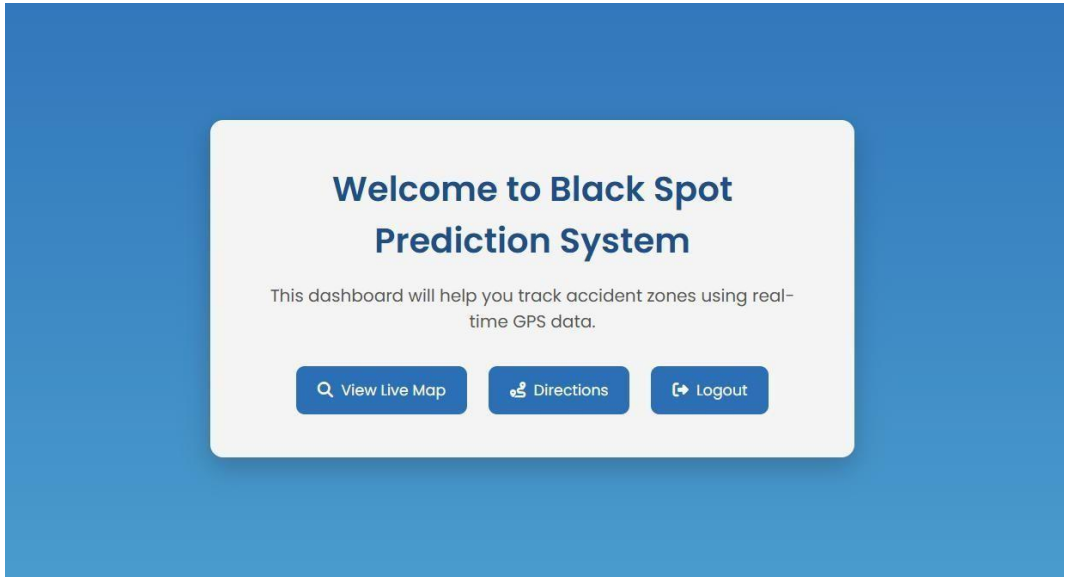


REGISTRATION PAGE

On clicking “Register Button” then it redirects to the registration page and opens interface like this and user enter their personal details (OTP facility also Included).

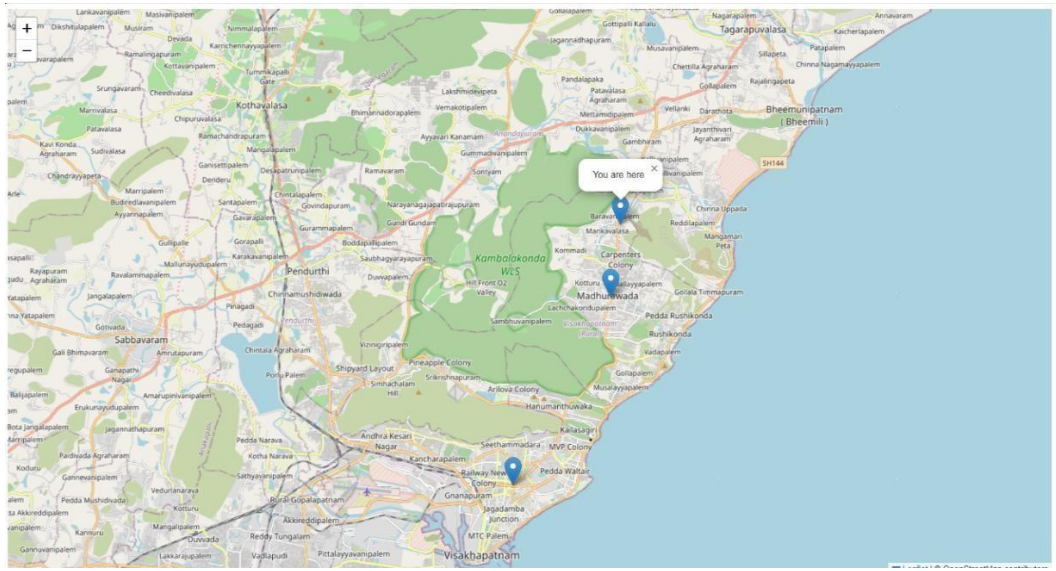
MODULE – 3 :

Module Status: Completed



DASHBOARD PAGE

Here is the “dashboard Page” enters after Login page which shows the Menu of the project

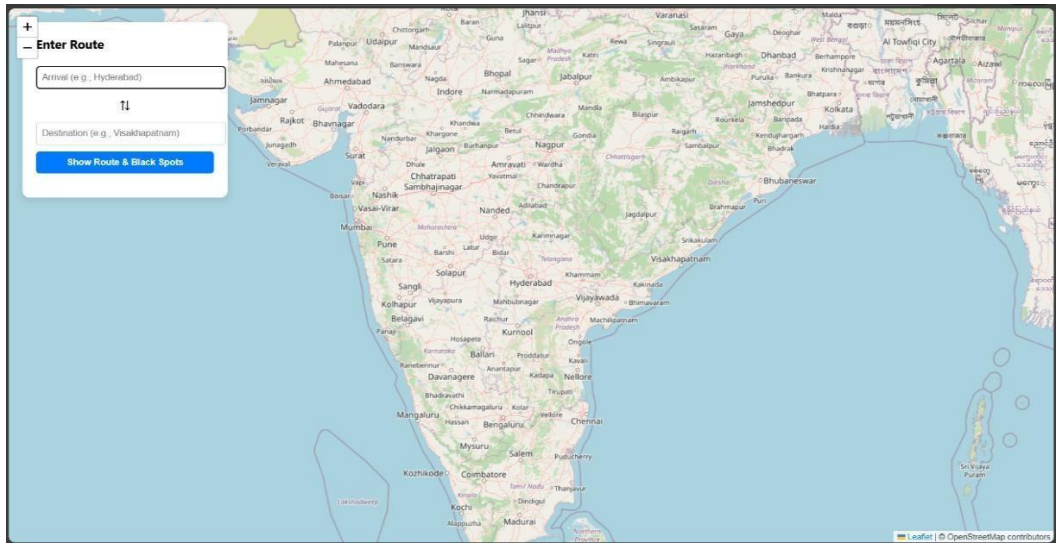


LIVE LOCATION PAGE

After clicking “Live Location Button” on the dashboard page we go through the Live location directly through Maps, user can proceed further.

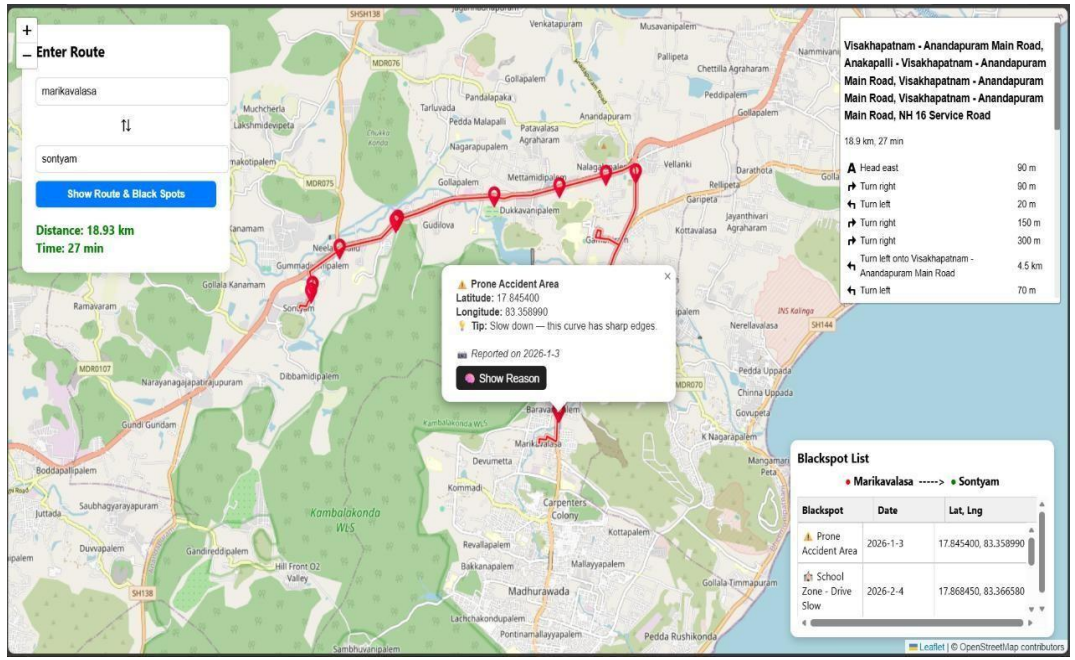
MODULE – 4 :

Module Status: Completed



Source & Destination page

The User click on the “directions” button on previous page will go through this page and enter the source & destination then it will get the result



Result page

After enter the Source & Destination address then it gives the result of directions, Blackspots in that direction and shows the route map. Shown in the Above Figure

MODULE – 5 :

Module Status: Completed

Blackspot List

● Marikavalasa -----> ● Sontyam

Blackspot	Date	Lat, Lng
⚠️ Prone Accident Area	2026-1-3	17.845400, 83.358990
🏫 School Zone - Drive Slow	2026-2-4	17.868450, 83.366580

Blackspot List Table

When click on the “Show Blackspot” Button then it will give you the list of the Blackspots while in the Route.

Conclusion

The Black Spot Detection System using Machine Learning successfully demonstrates how data-driven techniques can help identify accident-prone locations. By using geographical coordinates and a Random Forest model, the system provides quick and reliable predictions through a simple user interface.

The project shows that machine learning can support road safety planning by helping authorities focus on high-risk areas and take preventive measures. Although the current model uses limited features, it provides good accuracy and serves as a strong foundation for future improvements such as adding real accident data, traffic density, and road conditions.

Overall, the system proves that machine learning can play an important role in reducing road accidents and improving public safety.

FUTURE SCOPE

- Integrate real-time traffic and weather data.
- Use Deep Learning for higher prediction accuracy.
- Develop Android/iOS mobile applications.
- Connect with IoT sensors and smart traffic systems.
- Enable cloud-based storage and live monitoring.

REFERENCES

Python Documentation
Purpose: Programming language used for development.
Link: https://docs.python.org/3/

Scikit-learn
Purpose: Machine Learning algorithms and model training.
Link: https://scikit-learn.org/

Pandas
Purpose: Data preprocessing and analysis.
Link: https://pandas.pydata.org/

NumPy
Purpose: Numerical computations and array operations.
Link: https://numpy.org/

OpenStreetMap
Purpose: GPS map visualization and location mapping.
Link: https://www.openstreetmap.org/

Thank You